

Zixiang Wang

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Homepage: <https://fieldry.github.io/>

Google Scholar: 333 citations

Research Interest

I am a master's student at Peking University, advised by Prof. Liantao Ma. My research focuses on AI for Healthcare, especially medical large language models, electronic health record modeling, multi-agent collaboration, and reusable benchmark/toolkit infrastructure for clinical data analysis.

- (1) **Medical Large Language Models:** Building and evaluating LLM-driven systems for clinical prediction, reasoning, retrieval, and decision support.
- (2) **Electronic Health Record Modeling:** Modeling longitudinal, sparse, and multimodal EHR data with attention to clinical reliability and reuse.
- (3) **Benchmarks, Toolkits, and Protocols:** Creating open resources that make healthcare AI evaluation more reproducible and practically useful.

Education

Peking University 北京大学

2024.09 - Present

School of Software and Microelectronics · M.Eng. in Software Engineering (Working on AI for Healthcare, medical large language models, and EHR modeling under the supervision of Prof. Liantao Ma.)

Beihang University 北京航空航天大学

2019.09 - 2024.06

School of Software · B.Eng. in Software Engineering (Completed systematic training in software engineering and gradually shifted toward data modeling, machine learning, and healthcare AI research.)

Publications (* indicates equal contribution; † indicates corresponding author)

LLM Agents for Healthcare

- [WWW 2025] **ColaCare: Enhancing Electronic Health Record Modeling through Large Language Model-Driven Multi-Agent Collaboration**
Zixiang Wang*, Yinghao Zhu*, Huiya Zhao*, Xiaochen Zheng, Tianlong Wang, Wen Tang, Yasha Wang†, Chengwei Pan, Ewen Harrison, Junyi Gao†, Liantao Ma†
ACM International World Wide Web Conference (WWW), 2025
- [Preprint 2025] **HealthFlow: A Self-Evolving AI Agent with Meta Planning for Autonomous Healthcare Research**
Yinghao Zhu*, Yifan Qi*, Zixiang Wang, Lei Gu, Dehao Sui, Haoran Hu, Xichen Zhang, Ziyi He, Junjun He, Liantao Ma†, Lequan Yu†
Preprint, 2025
- [KDD 2024] **EHRFlow: A Large Language Model-Driven Iterative Multi-Agent Electronic Health Record Data Analysis Workflow**
Hao Wu*, Yinghao Zhu*, Zixiang Wang, Xiaochen Zheng, Ling Wang, Wen Tang, Yasha Wang, Chengwei Pan, Ewen Harrison, Junyi Gao†, Liantao Ma†
Artificial Intelligence and Data Science for Healthcare: Bridging Data-Centric AI and People-Centric Healthcare (KDD 2024 AIDSH Workshop), Oral, 2024

LLM for Healthcare

- [CIKM 2024] **EMERGE: Enhancing Multimodal Electronic Health Records Predictive Modeling with Retrieval-Augmented Generation**
Yinghao Zhu*, Changyu Ren*, **Zixiang Wang**, Xiaochen Zheng, Shiyun Xie, Junlan Feng, Xi Zhu, Zhoujun Li, Liantao Ma, Chengwei Pan[†]
ACM International Conference on Information and Knowledge Management (CIKM), 2024
- [Preprint 2024] **Prompting Large Language Models for Zero-Shot Clinical Prediction with Structured Longitudinal Electronic Health Record Data**
Yinghao Zhu*, **Zixiang Wang***, Junyi Gao, Yuning Tong, Jingkun An, Weibin Liao, Ewen Harrison, Liantao Ma[†], Chengwei Pan[†]
Preprint, 2024

Healthcare Benchmark

- [Preprint 2026] **Auditing medical multi-agent AI reveals risks of false consensus**
Yinghao Zhu*, Lei Gu*, **Zixiang Wang***, Haoran Sang, Dehao Sui, Wen Tang, Lan Mi, Yasha Wang, Junyi Gao, Liang Yao, Tianfan Fu, Ewen Harrison, Lequan Yu[†], Liantao Ma[†]
Preprint, 2026
- [npj Digital Medicine 2026] **ClinicRealm: Re-evaluating Large Language Models with Conventional Machine Learning for Non-Generative Clinical Prediction Tasks**
Yinghao Zhu*, Junyi Gao*, **Zixiang Wang***, Weibin Liao*, Xiaochen Zheng, Lifang Liang, Miguel O. Bernabeu, Yasha Wang, Lequan Yu, Chengwei Pan[†], Ewen Harrison[†], Liantao Ma[†]
npj Digital Medicine, 2026
- [NeurIPS 2025] **MedAgentBoard: Benchmarking Multi-Agent Collaboration with Conventional Methods for Diverse Medical Tasks**
Yinghao Zhu*, Ziyi He*, Haoran Hu*, Xiaochen Zheng*, Xichen Zhang, **Zixiang Wang**, Junyi Gao, Liantao Ma[†], Lequan Yu[†]
Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2025
- [Cell Patterns 2024] **A Comprehensive Benchmark For COVID-19 Predictive Modeling Using Electronic Health Records in Intensive Care**
Junyi Gao*, Yinghao Zhu*, Wenqing Wang*, **Zixiang Wang**, Guiying Dong, Wen Tang, Hao Wang, Yasha Wang, Ewen Harrison, Liantao Ma[†]
Cell Patterns, 2024

Healthcare Modeling

- [KDD 2025] **Learnable Prompt as Pseudo-Imputation: Rethinking the Necessity of Traditional EHR Data Imputation in Downstream Clinical Prediction**
Weibin Liao, Yinghao Zhu, Zhongji Zhang, Yuhang Wang, **Zixiang Wang**, Xu Chu, Yasha Wang, Liantao Ma[†]
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025
- [CIKM 2024] **PRISM: Mitigating EHR Data Sparsity via Learning from Missing Feature Calibrated Prototype Patient Representations**
Yinghao Zhu, **Zixiang Wang**, Long He, Shiyun Xie, Xiaochen Zheng, Liantao Ma[†], Chengwei Pan[†]
ACM International Conference on Information and Knowledge Management (CIKM), 2024

Trustworthy AI

- [KDD 2024] **RetCare: Towards Interpretable Clinical Decision Making through LLM-Driven Medical Knowledge Retrieval**
Zixiang Wang*, Yinghao Zhu*, Junyi Gao, Xiaochen Zheng, Yuhui Zeng, Yifan He, Bowen Jiang, Wen Tang, Ewen Harrison, Chengwei Pan, Liantao Ma[†], Ling Wang[†]
Artificial Intelligence and Data Science for Healthcare: Bridging Data-Centric AI and People-Centric Healthcare (KDD 2024 AIDSH Workshop), 2024

Toolkits & Platforms

- [CHI 2026] **Augmenting Clinical Decision-Making with an Interactive and Interpretable AI Copilot: A Real-World User Study with Clinicians in Nephrology and Obstetrics**
Yinghao Zhu*, Dehao Sui*, **Zixiang Wang***, Xuning Hu, Lei Gu, Yifan Qi, Tianchen Wu, Ling Wang, Yuan Wei, Wen Tang, Zhihan Cui, Yasha Wang, Lequan Yu, Ewen Harrison, Junyi Gao, Liantao Ma[†]
ACM CHI Conference on Human Factors in Computing Systems (CHI), 2026
- [KDD 2026] **OneEHR: Reproducible and AI Agent-Ready Longitudinal EHR Analysis Toolkit**
Yinghao Zhu, **Zixiang Wang**, Lei Gu, Dehao Sui, Yasha Wang, Ewen Harrison, Tianfan Fu, Junyi Gao, Lequan Yu[†], Liantao Ma[†]
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD-26) Hands-On Tutorials, 2026
- [STAR Protocols 2025] **Protocol for Processing Multivariate Time-series Electronic Health Records of COVID-19 Patients**
Zixiang Wang*, Yinghao Zhu*, Dehao Sui, Tianlong Wang, Yuntao Zhang, Yasha Wang, Chengwei Pan, Junyi Gao[†], Liantao Ma[†], Ling Wang[†], Xiaoyun Zhang[†]
STAR Protocols, 2025

[SAIL 2025]	AICare: An AI-Clinician Interaction System for Transparent and Actionable Clinical Decision Support Junyi Gao, Zixiang Wang , Dehao Sui, Wen Tang, Yasha Wang, Ling Wang, Liantao Ma, Yinghao Zhu <i>The Symposium on Artificial Intelligence in Learning Health Systems (SAIL), Best abstract nominee, 2025</i>
[STAR Protocols 2024]	Protocol to process follow-up electronic medical records of peritoneal dialysis patients to train AI models Tianlong Wang*, Yinghao Zhu*, Zixiang Wang , Wen Tang [†] , Xinju Zhao, Tao Wang, Yasha Wang, Junyi Gao [†] , Liantao Ma [†] , Ling Wang [†] <i>STAR Protocols, 2024</i>

Selected Project

▶ 小雅医生 (AICare): AI-Powered Clinical Decision Support Platform for Electronic Health Record Analysis, Deployed on PKU Xplore "clinical medicine + X" Platform	2025
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Selected Honors and Awards

▶ Outstanding Graduate of Beihang University, Beihang University	2024
▶ National Encouragement Scholarship, Beihang University	2022, 2023

Services

▶ Reviewer, AAAI 2026, 2027	2025, 2026
▶ Reviewer, npj Digital Medicine	2026
▶ Reviewer, npj Health Systems	2026
▶ Reviewer, Scientific Reports	2026